Uni-DAD: Unified Distillation and Adaptation of Diffusion Models for Few-step Few-shot Image Generation

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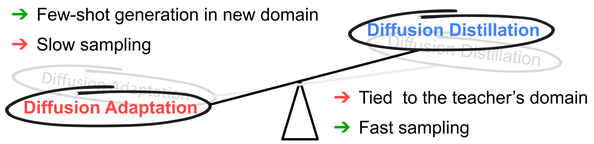
Motivation

Goal

Given a few (≤ 10) target images, produce high-quality and diverse target-domain samples with a few (1 – 4) steps.

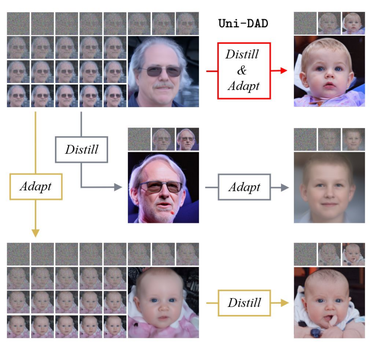
Distillation vs. Adaptation

- ➔ Few-shot generation in new domain
- ➔ Slow sampling



- ➔ Tied to the teacher's domain
- ➔ Fast sampling

Two-stage vs. Single-stage Distillation and Adaptation



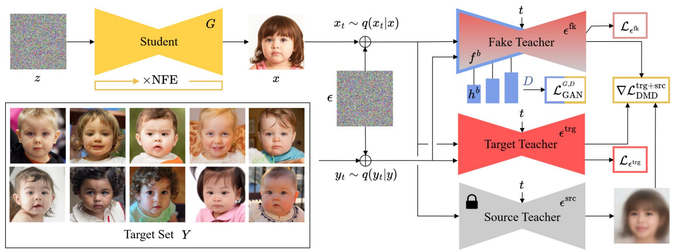
- ➔ Single-stage
- ➔ High-quality
- ➔ Diverse
- ➔ Two-stage
- ➔ Oversmoothing
- ➔ Two-stage
- ➔ Overfitting

Main Contributions

- ➔ We introduce the **first single-stage pipeline** that unifies distillation and adaptation of diffusion models.
- ➔ Dual-domain DMD and multi-head GAN are proposed to **retain source-domain diversity** while **sharpening target-domain realism** in few-shot settings.
- ➔ We **match or improve quality with far fewer steps** than prior adaptation methods, often surpassing two-stage pipelines in both quality and diversity.

Method

Uni-DAD Overview



Training Iteration

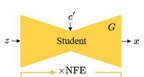
1 - Student update

Dual-domain DMD guides student toward source and target teachers. Multi-head GAN generator enforces realism.

Distribution Matching Distillation (DMD) $\nabla_\theta \mathcal{L}_{\text{DMD}^{\text{src/trg}}} \approx \mathbb{E}_{t,z}\left[\omega_t \left(\epsilon^{\text{fk}}(x_t) - \epsilon^{\text{src/trg}}(x_t)\right) \frac{dG_\theta}{d\theta}\right]$ (Fake Teacher; Source/Target Teacher)

Dual-domain DMD $\nabla_\theta \mathcal{L}_{\text{DMD}}^{\text{trg+src}} = (1-a)\nabla_\theta \mathcal{L}_{\text{DMD}^{\text{src}}} + a\nabla_\theta \mathcal{L}_{\text{DMD}^{\text{trg}}}$ (Dual-domain DMD Weight)

Multi-head GAN Generator $\mathcal{L}_{\text{GAN}}^{G}(\theta) = -\mathbb{E}_{t,z}\sum_{b\in\mathcal{B}} \log\left(h_\psi^b(f_\phi^b(x_t))\right)$

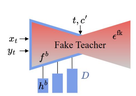


Student $\mathcal{L}_G(\theta) = \mathcal{L}_{\text{DMD}}^{\text{trg+src}}(\theta) + \lambda_{\text{GAN}}^G \mathcal{L}_{\text{GAN}}^G(\theta)$

2. Fake teacher + Discriminator update

Fake Teacher tracks student distribution, Multi-head GAN discriminator enforces realism.

Multi-head GAN Discriminator (D) $\mathcal{L}_{\text{GAN}}^D(\psi,\phi) = -\mathbb{E}_{t,y}\sum_{b\in\mathcal{B}} \log\left(h_\psi^b(f_\phi^b(y_t))\right) - \mathbb{E}_{t,z}\sum_{b\in\mathcal{B}} \log\left(1 - h_\psi^b(f_\phi^b(x_t))\right)$



Fake Teacher (fk) $\mathcal{L}_{\text{fk}}(\phi) = \mathbb{E}_{t,z}\left[\left\|\epsilon^{\text{fk}}{}_\phi(x_t) - \epsilon\right\|_2^2\right]$

fk+D $\mathcal{L}_{\text{fk}+D}(\phi,\psi) = \mathcal{L}_{\text{fk}}(\phi) + \lambda_{\text{GAN}}^D \mathcal{L}_{\text{GAN}}^D(\psi,\phi)$

3.Target teacher update (Optional):

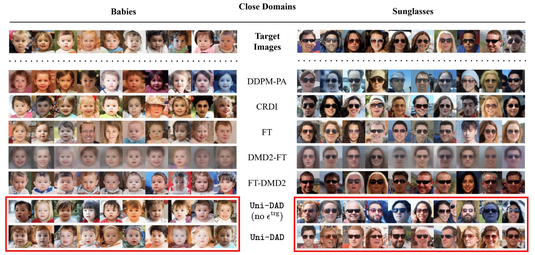
Improves adaptation when the target domain is structurally distant from the source.

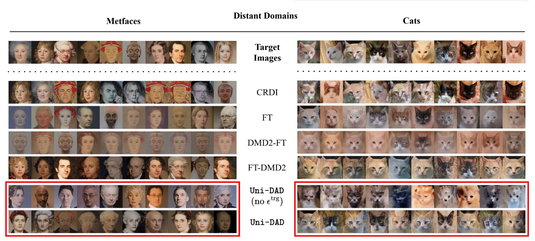
Target Teacher $\mathcal{L}_{\text{trg}}(\eta) = \mathbb{E}_{t,\epsilon,y}\left[\left\|\epsilon_\eta^{\text{trg}}(y_t) - \epsilon\right\|_2^2\right]$

Results

Few-shot Image Generation (FSIG)

- ➔ ≤ 4 NFE
- ➔ Improved SoTA
- ➔ Often surpasses two-stage in quality and diversity

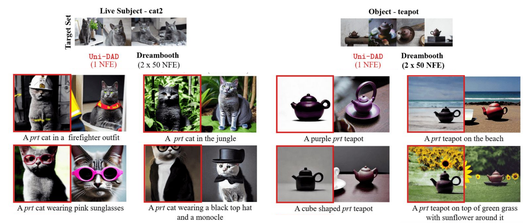


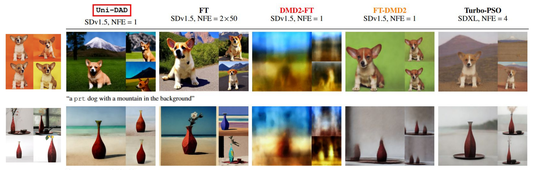


Variant	Method	NFE↓	1-stage	FID ↓ Babies	Sunglasses	MetFaces	Cats
Non-distilled	DDPM-PA [51]	1000	✓	48.92	34.75	—	—
	CRDI [3]	25	✓	48.52	24.62	121.36†	220.95
	FT	25	✓	57.06	37.86	72.99	61.62
Distilled	DMD2-FT	**3**		140.27	77.49	129.26	89.32
	FT-DMD2	**3**		57.11	41.85	63.25	**51.85**
	Uni-DAD (no ϵ^{trg})	**3**	✓	47.38	**22.57**	72.18	199.91
	Uni-DAD	**3**	✓	**45.09**	24.45	**58.13**	55.32

Subject-driven Personalization (SDP)

- ➔ 1 NFE
- ➔ Best stability across distilled baselines
- ➔ Prompt alignment, subject fidelity, diversity trade-off

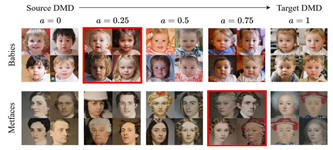




Method	NFE↓	1-stage	Quality DINO↑	CLIP-I↑	CLIP-T↑
FT [34]	2 × 50	✓	0.58	0.77	0.32
Turbo-PSO$_{\text{SDXL}}$ [27]	4		0.50	0.70	0.30
DMD2-PSO$_{\text{SDv1.5}}$ [27]	**1**		0.14	0.56	0.23
DMD2-FT	**1**		0.20	0.61	0.26
FT-DMD2	**1**		**0.57**	**0.75**	0.25
Uni-DAD	**1**	✓	0.47	0.73	**0.29**

DMD-domain DMD Weight

Can control target vs. source influence.



Robust to Different Steps and Shots

Constantly outperforms CRDI.

Methods	NFE↓	1-shot B	1-shot M	5-shot B	5-shot M	10-shot B	10-shot M
CRDI	25	105.51	145.10	51.71	126.34	48.52	121.36
Uni-DAD	4	**72.38**	95.44	**45.86**	**81.85**	**41.39**	59.49
Uni-DAD	3	90.33	**90.29**	52.73	83.69	45.09	**58.13**
Uni-DAD	2	95.69	114.78	68.75	98.36	62.45	79.08
Uni-DAD	1	109.55	132.79	93.52	103.84	98.52	89.03

Adapting or Distilling or Both?

Can initialize from pre-trained weights.

Available Checkpoints *Pre-distilled G*	*Pre-adapted ϵ^{trg}*	FID ↓ Babies	MetFaces
		45.09	58.13
✓		42.68	77.80
	✓	46.73	62.67
✓	✓	**42.04**	**54.01**